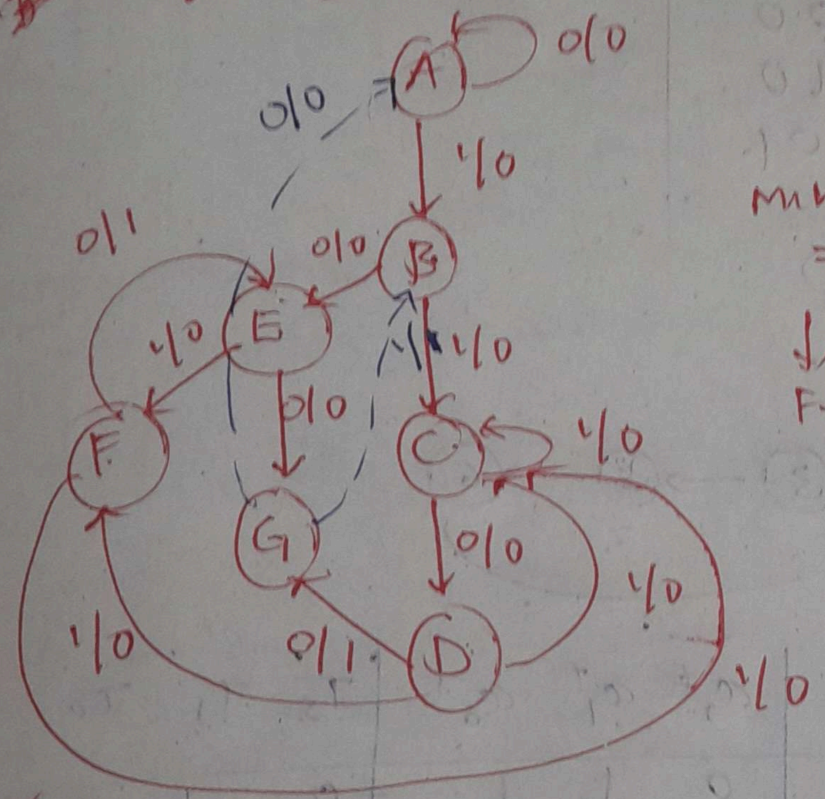
#2)

(a) Mealy (~~...~~ overlap) ✓

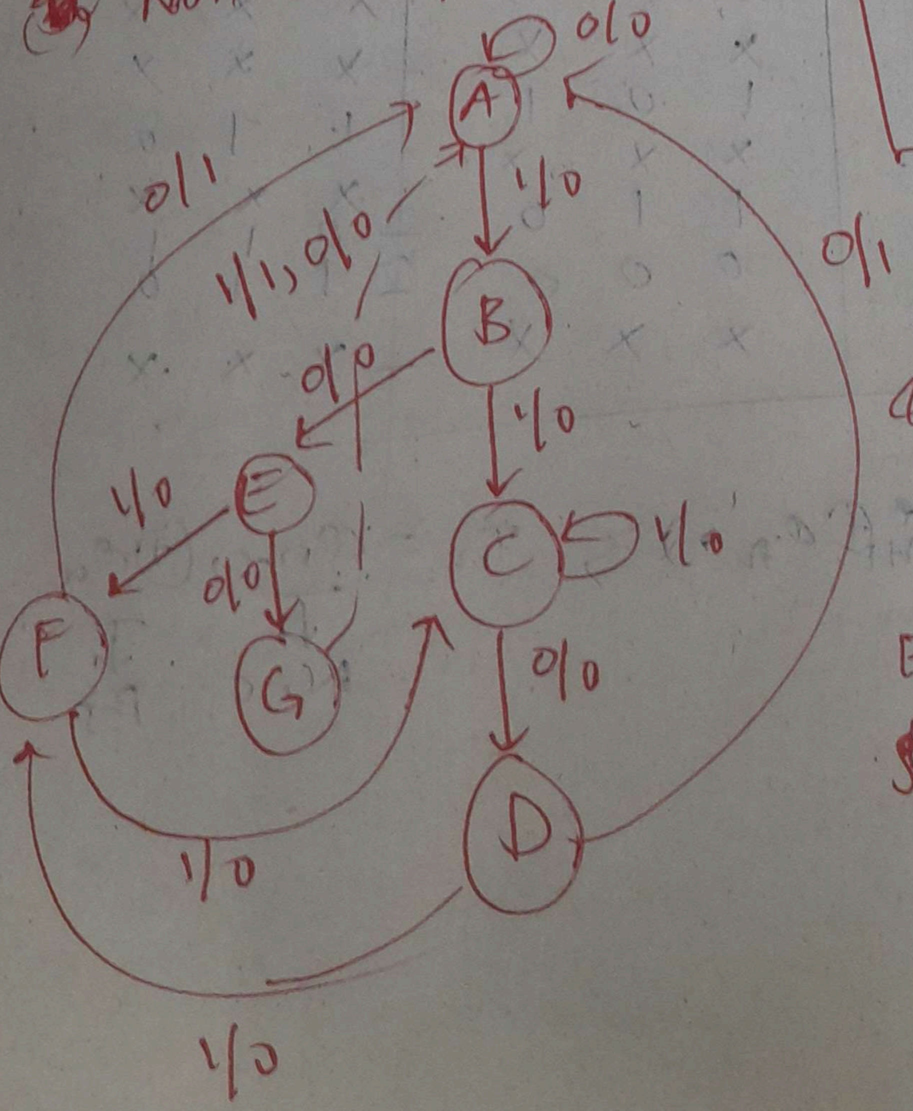


Min. state = 7

↓

F.F's = 3

(b) Non-overlap



a → 8 + 2

b → 8 + 2

Que

a → 8

b → 8

F.F's → 1

States → 1

#
(3) $0 \to 3 \to 5 \to 6 \to 0$

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	1	1	0	1	1
0	0	1	x	x	x	x	x	x
0	1	0	x	x	x	x	x	x
0	1	1	1	0	1	1	1	0
1	0	0	x	x	x	x	x	x
1	0	1	1	1	0	0	1	1
1	1	0	0	0	0	1	1	0
1	1	1	x	x	x	x	x	x

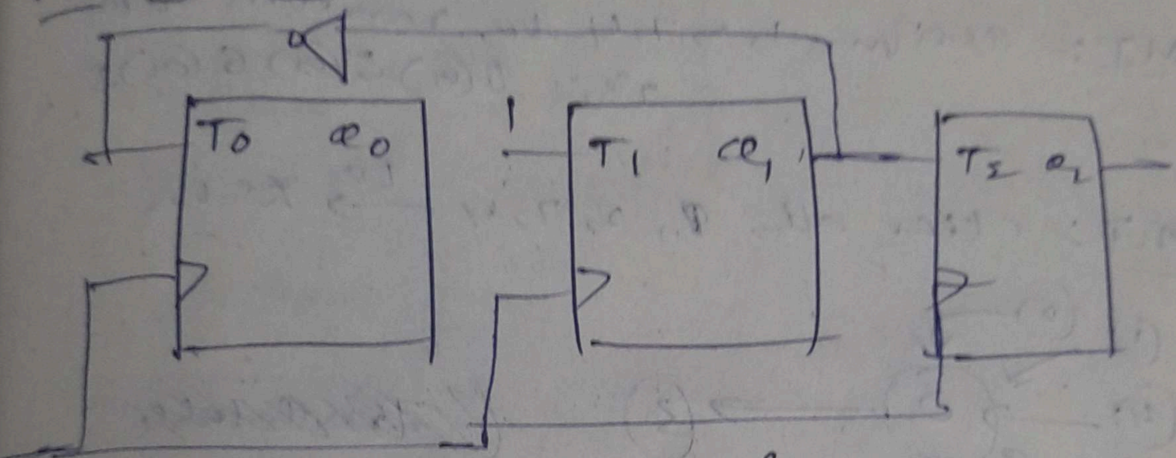
$T_n = Q_{n+1}^{+} \oplus Q_n$, $T_n = \underbrace{Q_{n+1}}_{N.S} \oplus \underbrace{Q_n}_{P.S}$

T_2	$\bar{Q}_1\bar{Q}_0$	$\bar{Q}_1 Q_0$	$Q_1 Q_0$	$Q_1\bar{Q}_0$
$\bar{Q}_2$	0	x	1	x
Q_2	x	0	x	1

$= Q_1$

$||^{ly}$, $T_1 = 1$; $T_0 = \bar{Q}_1$

Design of lockout:



clk

$(\because T_n = Q_{n+1} \oplus Q_n$

$\Rightarrow \boxed{Q_{n+1} = T_n \oplus Q_n}$

Now, states 1, 2, 4, 7

Q_2 Q_1 Q_0	Q_2^+	Q_1^+	Q_0
0 0 1	0	1	0
0 1 0	1	0	0
1 0 0	1	1	1
1 1 1	0	0	1

T_n is known

$T_2 = Q_1$

$T_1 = 1$

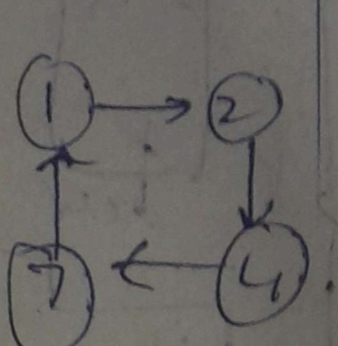
$T_0 = \bar{Q}_1$

$Q_2^+ = Q_1 \oplus Q_2$

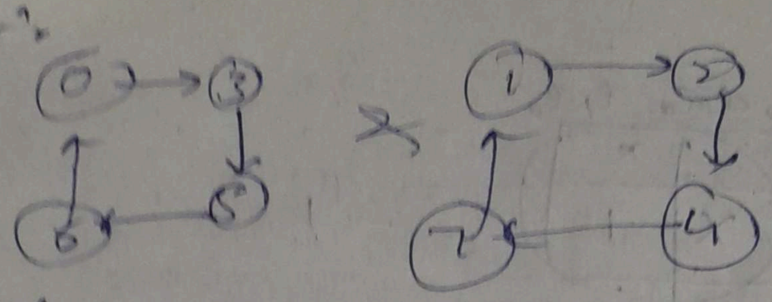
$Q_1^+ = 1 \oplus Q_1 = \bar{Q}_1$

$Q_0^+ = \bar{Q}_1 \oplus Q_0 = Q_1 \odot Q_0$

$\Downarrow$



Ex

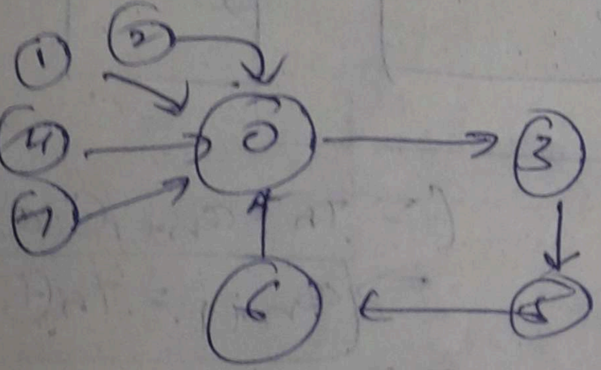


lockout!!, Yes

→ Avoiding technique:

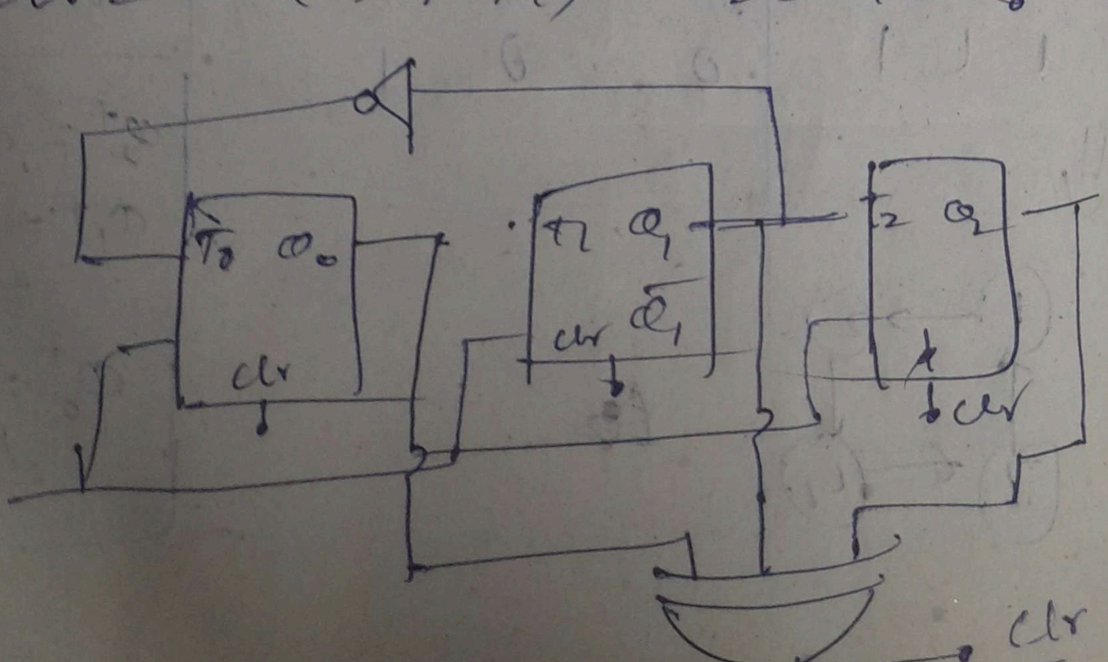
MI :- assign 1, 2, 7, 4 to random states
i.e., 0 (or) 3 (or) 6 (or) 5.

MII :- clear all 1, 2, 7, 4 $\xrightarrow{\text{to}}$ zero



Ans :-

$Clr = \Sigma m(1,2,4,7) = Q_2 \oplus Q_1 \oplus Q_0$



#
(5)

From In to (D_2)

$$\boxed{T_{clk}' \geq T_{clq_1} + T_{dly_1} + (T_{su_2}) - T_{skew_1} \\ \& \; T_{h_2} \leq T_{clq_1} + T_{dly_1} - T_{skew_1}}$$

from $I_{n D_2}$ to D_3.

$$T_{clk}'' \geq (T_{clq_2} + \cancel{T_{skew_1}}) + T_{dly_2} + T_{su_3} - (\cancel{T_{skew_1}} + T_{skew_2})$$

$$\geq T_{clq_2} + T_{dly_2} + T_{su_3} - T_{skew_2}$$

$$\& \; \boxed{T_{h_3} \leq T_{clq_2} + T_{dly_2} - T_{skew_2}}$$

$$\boxed{f_{max} = \frac{1}{T_{min}}}$$

$\therefore T_{min} = ? = max(min(T_{clk}'), min(T_{clk}''))$

This $Q=1$ goes to G_7 & G_6

$G_{7out} = (Q)(K')(Y)$; $G_6 = 0$ ($\because Clk = 0$)

$= (1)(1)(1) = 1$

$\Rightarrow G_{8o} = \boxed{0 = \bar{Q}}$

In region (7)

when $Y=0$

$\Rightarrow G_7 = G_6 = 0$ immediately $\Rightarrow \boxed{G_8 = 1 = \bar{Q}}$

this $\bar{Q}$ goes to G_3 & G_2

$G_3 = (\bar{Q})(J')(X) = (1)(1)(1) = 1$

$G_2 = (X)(Clk)(\bar{Q}) = (1)(0)(1) = 0$

$\Rightarrow G_4 = \overline{1+0} = \boxed{0 = Q}$

After $Y=1$

still $Clk = 0$ & $J' = K' = 1$

$G_3 = (\bar{Q})(J')(X) = 1$

$G_2 = (Q)(Clk)(X) = 0$ } $G_4 = \boxed{0 = Q}$

$\Rightarrow G_7 = 0 = G_6$ (due to Q)

$\Rightarrow \boxed{\bar{Q} = 1}$

$Y =$

$\because$ As $X = 0$, $Q = 1$

& for $Y = 0$; $Q = 0$

(1C) The circuit is a negative triggered flipflop as output state (Q) changes only at negative edge of clock.

(4)

Clock	Q_A	Q_B	Q_C	Q'_A	Q'_B	Q'_C	X'	X''	X'''	Z
	1	0	0	1	0	0	0	0	0	0
1	0	1	0	1	1	0	1	0	0	1
2	0	0	1	1	1	1	1	1	0	1
3	1	0	0	0	1	1	1	1	1	1
4	0	1	0	0	0	1	0	1	1	1
5	0	0	1	0	0	0	0	0	1	1
6	1	0	0	1	0	0	0	0	0	0

$\therefore$ 6 clock pulses to $\boxed{Z=0}$

after 't_d', $K' = 0$

$G_{10} = 0$

$G_{6o} = (clk) \cdot (Y) \cdot (Q)$

$= (1) \cdot (1) \cdot (1)$

$= 1$

$\therefore G_{8o} = \boxed{\bar{Q} = \overline{1+0} = 0}$ still & this $\bar{Q}$ goes to

G_3 & G_2 $= 0 \therefore \boxed{Q = \bar{0} = 1}$

at region ③ ie when clock goes -ve

$K' = 1$ but with delay. so

$G_7 = (K'_{\text{bef delay}}) \cdot Q \cdot Y = 0$

$G_6 = clk = 0 \therefore = 0$

$\Rightarrow G_8 = \overline{G_7 + G_6} = \bar{0} = \boxed{1 = \bar{Q}}$

this goes to G_3 & G_2

$G_3 = (\bar{Q} \cdot J' \cdot X)$ as $clk = 0$ ($J' = 1$ still)

$= (1) \cdot (1) \cdot (1) = 1$ ($\because$ NOR atleast one '1' o/p = 0)

$\Rightarrow \boxed{G_{4o} = 0 = Q}$

So this continues till region 4 as no changes in i/p

at region 5

$clk = 1$ but no change in $G_7 = G_6 = 0 \Rightarrow \bar{Q} = 1$, $G_3 = 1 \Rightarrow Q = 0$

$J' = \overline{(clk) \cdot (B) \cdot \bar{Q} \cdot Y}$

$= \overline{(1) \cdot (0) \cdot (1) \cdot (1)} = 1$ still

$\therefore$ clk alone has no effect on J'

now once ~~B=0~~ 'B = 1'

$J' = \overline{(1) \cdot (1) \cdot (1) \cdot (1)} = \bar{1} = 0$ (but with delay)

after delay,

$G_3 = (\bar{Q} \cdot J' \cdot X) = (1)(0) \cdot (1) = 0$

$G_2 = (\bar{Q} \cdot clk \cdot X) = 1 \Rightarrow G_{4out} = \boxed{0 = Q}$

this 0 goes to G_7 & G_6 making $\boxed{\bar{Q} = 1}$

after this ~~[illegible]~~ $A = 0$

$\Rightarrow K' = 1$ previously $K' = 1 \therefore$ no change in K'

Now in region 6 : when $clk = 0$

$K' = 1$ so no change

but $J' = 1$ but with delay.

$G_{3out} = (J'_{\text{bef delay}}) \cdot (\bar{Q}) \cdot (X)$

$= 0$

$G_2 = 0$ ($\because clk = 0$)

$\Rightarrow \boxed{G_{4out} = 1 = Q}$

Assume O/ps after delay as J' & K' then

In ① region as $T/4 >> t_d$ so,

for G1

$x = 0 \Rightarrow K' = 1$

for G_5 $clk = 0$

$\Rightarrow J' = 1$

for G_3, G_2 as $x = 0 \rightarrow G_{3out} = G_{2out} = 0$

$\therefore \boxed{Q = 1}$

for $G_{7out} = Q \cdot K' \cdot y = (1)(1)(1) = 1$

$G_{6out} = 0$ ($\because clk = 0$)

$\rightarrow G_{8o} = \overline{1+0} = \boxed{0 = \overline{Q}}$

In region ②

when $A = 1$

still $x = 0 \therefore K' = 1$; ~~[illegible]~~ $\therefore J' = 1$

still $x = 0 \rightarrow G_3 = G_2 = 0 \rightarrow \boxed{Q = 1}$

$G_{7o} = Q \cdot K' \cdot Y = (1) \cdot (1) \cdot (1) = 1$

$G_{6out} = clk \cdot y \cdot Q = 1$

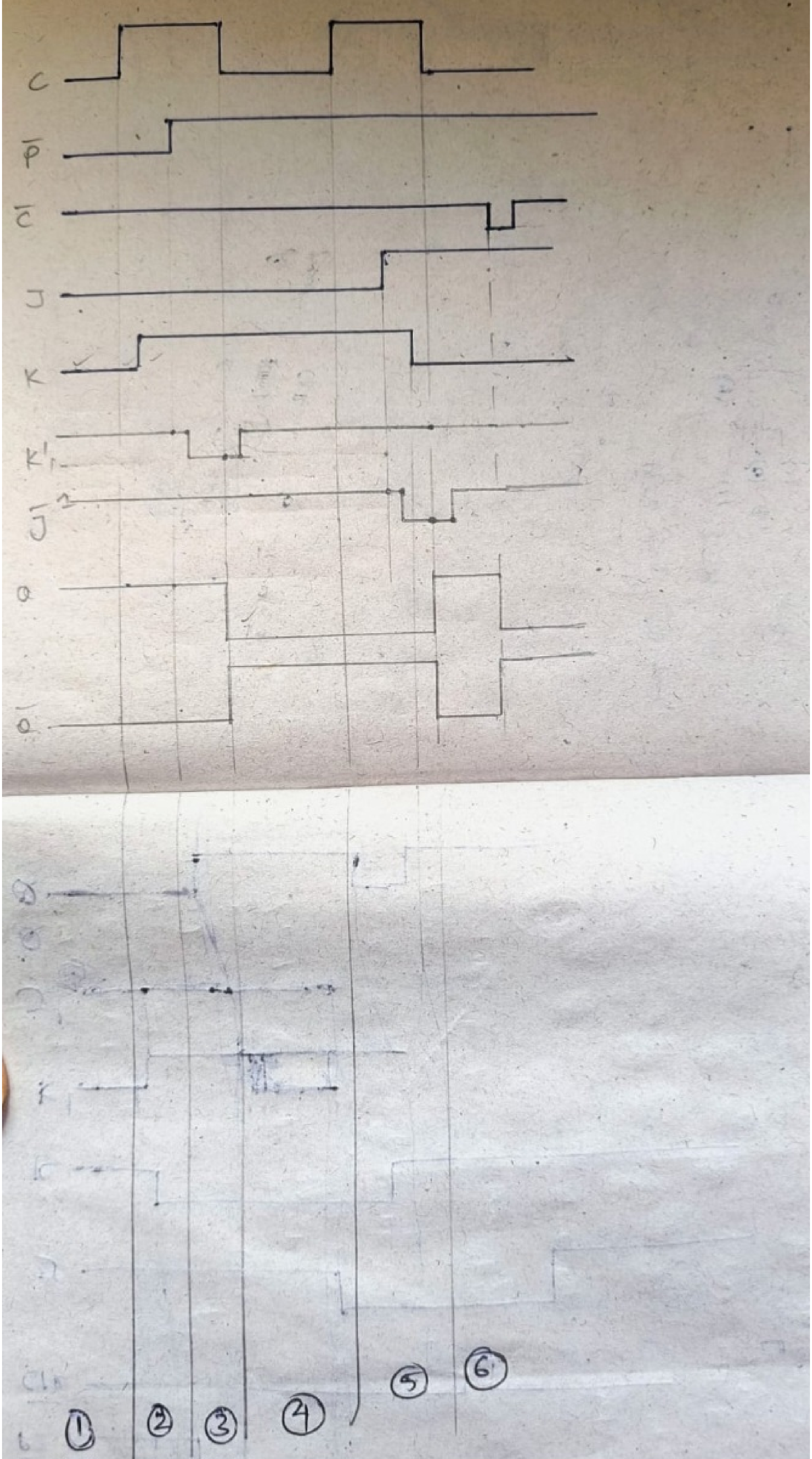
$G_{8out} = \overline{1+1} = \overline{1} = \boxed{0 = \overline{Q}}$

now when $x = 1$

$\Rightarrow K' = \overline{(Q) \cdot (x) \cdot (A) \cdot (clk)}$

$= \overline{(1) \cdot (1) \cdot (1) \cdot (1)}$

... delay t_d


C
$\bar{P}$
$\bar{C}$
J
K
K'
$\bar{J}$
Q
$\bar{Q}$
1
2
3
4
5
6